

Dr Andi Lowe

Data Scientist · PhD in Particle Physics · Ex-CERN · Breakthrough Prize Laureate

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Professional Profile

- Machine Learning-focused Data Scientist with 8+ years of industry experience delivering analytical and ML-driven solutions in regulated and technically complex environments. Background in Python and R development, performance optimisation, and algorithmic problem-solving, with experience spanning scientific computing, statistical modelling, large-scale data processing, and applied machine learning.
- PhD-qualified with extensive experience working with messy, real-world data and a strong focus on data quality assessment, feature engineering, and rigorous model validation. Proven consultant and stakeholder manager, adept at translating complex analytical results into clear, actionable insights for senior business stakeholders across diverse domains.
- Former CERN researcher and member of the team that discovered the Higgs boson.
- British citizen with unrestricted right to work in Ireland; immediately available to relocate.

Work History

Data Scientist (Client-Facing Consulting Role)

EPAM Systems Inc.

Budapest, Hungary

June 2017 – present

Led end-to-end data science and applied machine learning engagements for global clients, working independently across the full analytics lifecycle, from problem definition and modelling through validation, documentation, and delivery of decision-support solutions in business and operational settings.

- Developed, validated, and documented predictive models on large, noisy time-series datasets using AWS SageMaker, Python, and SQL to identify failure risk and operational anomalies in GPS fleet-tracking data. Delivered actionable insights to operations teams; work resulted in a patent filing (*Verizon Connect*).
- Built a proof of concept for remote sensing using millimetre-wave radar technology, performing signal analysis and image processing with OpenCV (*Texas Instruments*).
- Designed and implemented machine learning models in Python (scikit-learn) to detect clinically relevant behavioural patterns from wearable sensor data, supporting treatment monitoring by healthcare professionals. Achieved 87% average gesture-recognition accuracy (79–100% per gesture), validating feasibility and enabling progression to MVP. Provided technical leadership by directing data collection strategy and advising on hardware selection to align analytical and practical constraints (*Child Mind Institute*).
- Performed end-to-end analysis of neuromarketing data to identify key drivers and inform campaign strategy. Presented insights directly to senior business stakeholders (*Merck Sharp & Dohme*).
- Led the design and implementation of privacy-preserving data pipelines, ensuring GDPR compliance and protecting a critical ~\$250M annual revenue stream. Supported data governance in a highly regulated environment and provided evidence-based guidance on reducing digital carbon footprint (*London Stock Exchange Group*).
- Built extract, transform, and load (ETL) processes using KNIME to automate Excel financial report generation, reducing processing time from over 24 hours to under 2 hours (*Bayer AG*).

Research Fellow

Wigner Research Centre for Physics, Hungarian Academy of Sciences

Budapest, Hungary

Sept 2013 – May 2017

Conducted large-scale data analysis, statistical modelling, and machine learning in R and C++ for the ALICE experiment at CERN, which recreates conditions believed to have existed shortly after the Big Bang.

- Developed ML classification algorithms to identify particles based on their decay properties.
- Engaged with the data science community through public outreach talks and conference presentations.

Postdoctoral Fellow and Deputy Team Leader

California State University, Fresno

Fresno, CA, USA

Apr 2010 – Oct 2012

Investigated the utility of predictor variables for data-mining analyses using Monte Carlo simulations in C++, engineering features to improve search sensitivity for new particles and reduce the time and cost of experimental data collection.

- Contributed to National Science Foundation (NSF) grant proposals totalling over \$1.6 million in research funding.
- Stationed at CERN, Geneva, Switzerland.

Postdoctoral Fellow

Indiana University

Bloomington, IN, USA

Feb 2008 – Aug 2009

Developed high-performance algorithms in C++ and Python for real-time data processing at input rates of ~1 GB/s, optimized for the ATLAS experiment's physics programme and used in production to process tens of petabytes of data in a distributed scientific computing environment.

- Stationed at CERN, Geneva, Switzerland.

Education

Royal Holloway, University of London

Egham, UK

Doctorate (PhD) in Particle Physics

Designed and optimised high-performance C++ software for real-time data reduction pipelines, including profiling, memory optimisation, and scalability improvements in a large, long-lived codebase.

- Wrote software for a real-time multi-stage cascade classifier that reduces collision event data rates from 60 TB/s to 300 MB/s for permanent storage.
- Performed detailed time profiling of the core software and devised improvements that made it 8 times faster, thus meeting a critical requirement of the system.
- Analysed petabyte-scale datasets using distributed computing systems.
- Developed software used in the discovery of the Higgs boson.
- Stationed at CERN, Geneva, Switzerland for 1.5 years.

Royal Holloway, University of London

Egham, UK

Master's degree (MSc) in Particle Physics

Investigated the search potential for the Higgs boson via the H to bb (bottom quark-antiquark) decay channel using Monte Carlo simulations with the ATLAS detector at CERN.

- Conducted the first data-mining analysis of this type entirely in C++, setting a benchmark for subsequent research in the field.
- Achieved a signal significance of 4.87 sigma (previously 3.66 sigma; 5 sigma is the threshold for discovery).

Royal Holloway, University of London

Egham, UK

Bachelor's degree (BSc) in Physics

Leadership and Consulting Skills

Communication: Skilled at collaborating with clients as a consultant and effectively communicating with business stakeholders and technical teams to convey complex data-driven insights and technical specifications. Proficient in translating client requirements into detailed project documentation. Invited speaker at numerous international conferences. Experienced in visual storytelling and data visualisation best practices. Adept at simplifying complex technical concepts for diverse audiences. Extensive experience working with remote, distributed teams.

Problem solving: Proven ability to lead independent research, analyse complex problems, and develop innovative, effective solutions.

Project management: Experienced in managing multiple projects under strict deadlines, both onsite and remotely. Familiar with Agile and Waterfall software development methodologies.

Professional Certificates

- **AWS Certified AI Practitioner**, Amazon Web Services (AWS), issued Nov. 2024.
- **Salesforce Certified AI Associate**, Salesforce, issued Jan. 2025.

Computing Skills

Machine learning frameworks/tools: Python-based ML tooling (Scikit-Learn, XGBoost, Keras), AWS Sagemaker, KNIME, Weka, H2O, caret, ROOT (CERN's C++ data analysis framework).

Programming languages: Python (NumPy, Pandas, Numba, etc), R, C++, Bash, SQL, Octave.

Data visualisation: Matplotlib, Seaborn, Plotnine, ggplot2, Plotly, TIBCO Spotfire, Tableau, Flexdashboard.

Notebooks/Documentation: Jupyter, Google Colab, R Markdown, Confluence, LaTeX, Doxygen.

Software development: Docker, Git/GitHub/GitLab, Jira, RStudio, UML, Valgrind, Visual Studio Code.

Operating systems: Unix, Linux, Microsoft Windows.

Other: object-oriented analysis and design, CPU and time profiling, code optimisation, memory debugging.